

AI Receptionist

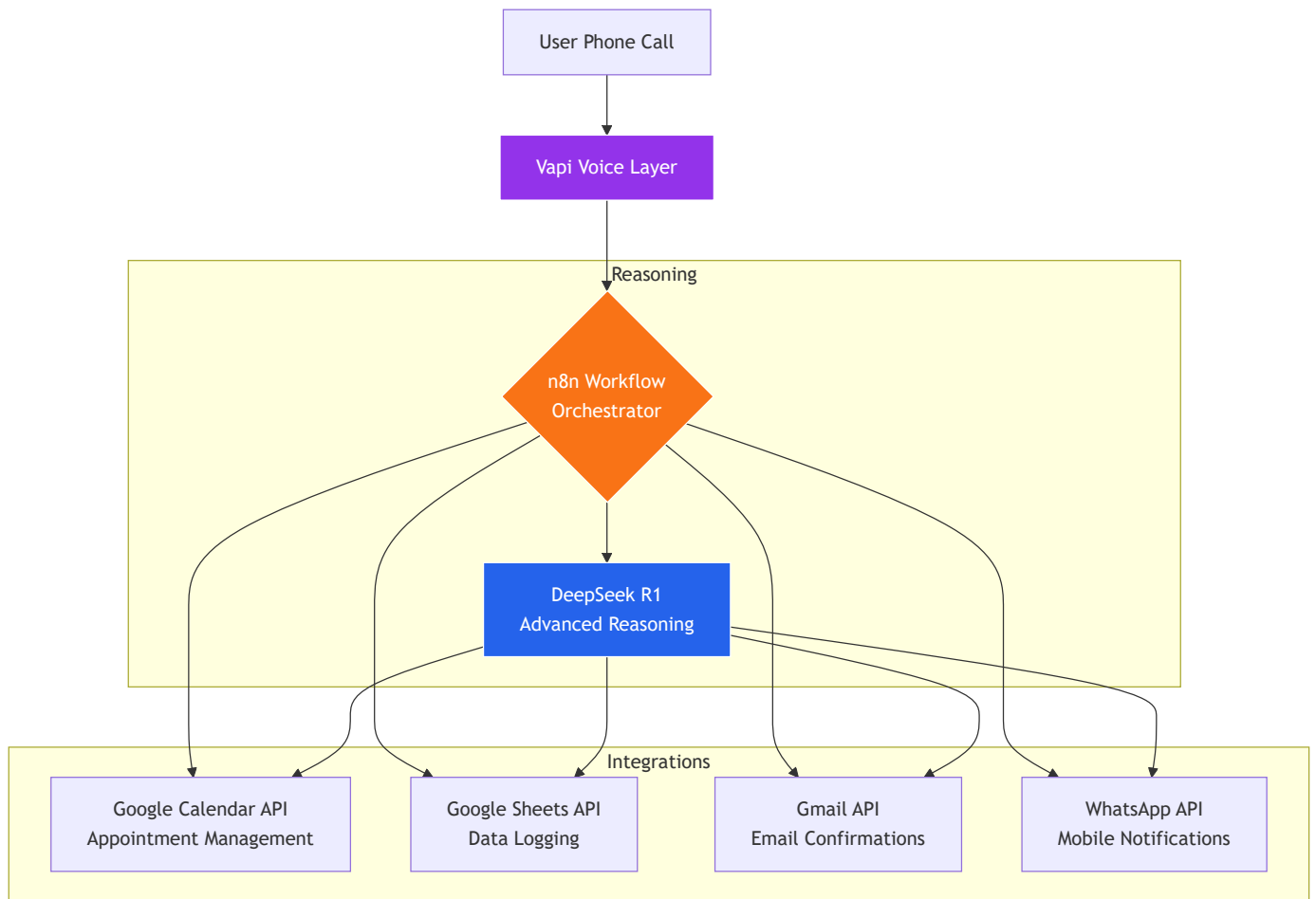


Executive Summary

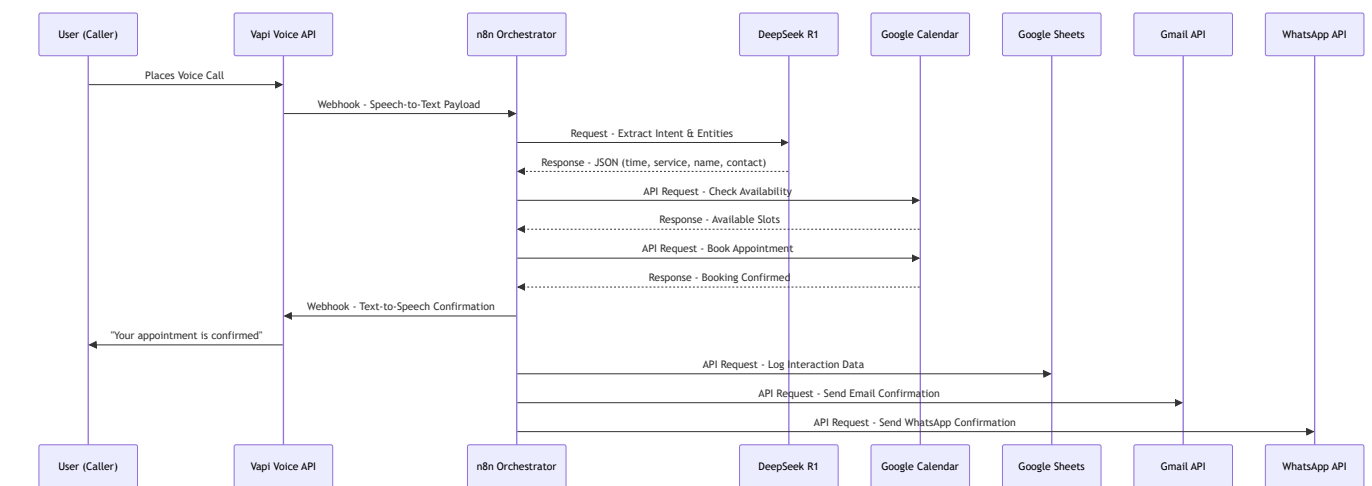
AI Receptionist is an intelligent voice-based virtual receptionist engineered to eliminate hold times and automate front-desk operations through human-like conversational AI. By seamlessly integrating real-time voice processing with dynamic scheduling capabilities, the system autonomously manages appointment bookings, rescheduling, and multi-channel confirmations. Developed in January 2026 by Amol Dadarao Rathod (B.Tech in AI & Data Science), this project propelled its team to the finals of the prestigious **OpenAI Academy x Nxt Wave Buildathon**, demonstrating production-ready automation at scale.

Deep-Dive System Architecture

High-Level Architecture Flow



Appointment Booking Sequence



Feature Highlights & Automation Workflows

Core Functionalities

- **Conversational Voice AI:** Handles inbound calls with <200ms latency, dynamically adapting to user queries using contextual understanding
- **Dynamic Real-Time Scheduling:** Searches, books, and reschedules appointments without human intervention, checking calendar availability in-flight
- **Intelligent Entity Extraction:** Parses caller data (name, contact, service type, preferred time) directly from natural language conversation
- **Multi-Channel Confirmations:** Simultaneous webhook orchestration triggers personalized confirmations via Email and WhatsApp within seconds of booking
- **Persistent Data Logging:** All call interactions stored in Google Sheets with timestamp, caller details, and appointment metadata

Edge Case Handling

- **Rescheduling Conflicts:** When conflicts arise, the system proposes 3 alternative time slots based on calendar analysis and solicits user preference
- **Ambiguous Time References:** Natural language processing interprets relative dates ("next Tuesday", "afternoon") and converts to structured datetime formats
- **Incomplete Information:** Context retention across conversation turns prompts users for missing data (contact info, service type) without redundant questions
- **API Failures:** Graceful fallback mechanisms queue retry attempts and notify system administrators via webhook alerts

n8n Webhook Orchestration Power

The n8n platform serves as the central nervous system, managing:

- Asynchronous webhook reception from Vapi's real-time streaming
- Parallel API execution to Google services and WhatsApp
- Conditional branching based on LLM response parsing
- Error state management with automatic retries
- Data transformation between JSON schemas and API payloads

Technical Stack Breakdown

Technology	Role	Key Integration Points
DeepSeek R1	Advanced Reasoning & NLP Engine	Intent classification, entity extraction, scheduling logic, conflict resolution
Vapi	Voice I/O & Telephony Infrastructure	Real-time speech-to-text streaming, text-to-speech synthesis, phone number provisioning, call session management
n8n	Workflow Orchestration & API Bridge	Webhook handling, REST API calls, JSON transformation, conditional logic, error handling
Google Calendar API	Appointment Management	Availability checking (<code>calendar.freebusy.query</code>), event creation (<code>calendar.events.insert</code>), real-time updates
Google Sheets API	Persistent Data Storage	Interaction logging via <code>sheets.spreadsheets.values.append</code> , customer database management
Gmail API	Email Confirmations	Message composition via <code>gmail.users.messages.send</code> , template rendering with appointment details
WhatsApp API	Mobile Confirmations	Instant messaging via webhook-triggered templates, real-time delivery status

Challenges Conquered & Future Scope

Technical Challenges

Latency Optimization in Voice AI: Achieving conversational-grade response times required implementing streaming speech-to-text with progressive webhook triggers, reducing perceived latency to under 300ms through predictive pre-fetching of common workflow branches and optimized prompt engineering for DeepSeek R1.

Structured JSON Output Engineering: Converting freeform conversational text into deterministic JSON payloads demanded custom prompt templates with constrained

output formatting, regex-based post-processing validation, and schema inference layers within n8n workflows to ensure API compatibility.

State Management at Scale: Managing conversation context across webhook boundaries required implementing a Redis-backed session store with TTL-based cleanup, ensuring continuity without memory leaks in long-running deployments.

Future Enhancements

- **Multi-Language Support:** Extend to regional language recognition and response generation
- **CRM Integration:** Connect Salesforce/HubSpot APIs for enterprise customer context
- **Voice Biometrics:** Implement speaker verification for enhanced security
- **Analytics Dashboard:** Real-time metrics on call volume, success rates, and user satisfaction
- **Scheduled Callbacks:** Automated follow-up appointment reminders via the same voice channel

*Project by Amol Dadarao Rathod | B.Tech in AI & Data Science | OpenAI Academy x
Nxt Wave Buildathon Finalist*